GEOHAZARD TECHNICAL SOLUTIONS (GTS)

Geotechnical Solutions for landslide-prone areas of Nepal

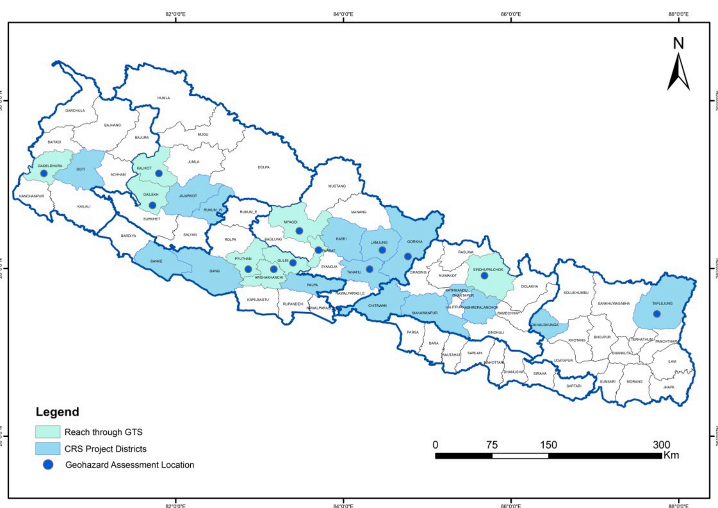
Nisdi Rural Municipality of Lumbini Province faced a large landslide triggered after heavy rainfall, putting 23 households at high risk. Photo: CRS Nepal

Background

Nepal's rugged terrain, intense monsoon rains, and seismic activity trigger frequent landslides, with 494 reported in 2023 (bipadportal.gov.np). Limited technical capacity hinders timely geohazard assessment across all tiers of government. Catholic Relief Services (CRS) Nepal and nodal government bodies such as the National Disaster Risk Reduction and Management Authority (NDRRMA) recognize the need of collaborative efforts to assess and mitigate landslide risks for vulnerable communities in high-risk areas.

Understanding this context and pressing need, CRS Nepal started the '**Geohazard Technical Solutions (GTS)**' project in 2023 to support Nepal government in identifying and mitigating geohazards.

GTS prioritizes fostering resilience by supporting at-risk communities, while also developing affordable and scalable disaster risk reduction approaches, simplifying guidance and tools for wider adoption.



Project Coverage Areas

CRS Nepal in Action

- **In-field geo-hazard assessment:** of more than **86** sites in **22** Local Governments (LGs) across **5** provinces (Koshi, Bagmati, Gandaki, Lumbini, Karnali and Sudurpaschim). The process for relocation, reconstruction and mitigation has started in most of the assessment sites.
- **Guideline development: The Geohazard Assessment Guideline 2080** has been drafted in coordination with NDRRMA, National Association of Rural Municipalities in Nepal (NARMIN), Department of Mines and Geology and Center for Disaster Studies (Institute of Engineering, Tribhuvan University). It enables municipal engineers to conduct timely assessments, emphasizing capacity building and local resource utilization for informed decision-making.
- **Leveraged domestic funds for bio-engineering:** The GTS project supported two LGs of Gandaki Province (Marshyangdi and Mangala Rural Municipality) to adopt rural road construction guidelines aiming to reduce landslide risks with bioengineering, recommended gradient and drainage management. NPR 1 million has been allocated by Marshyangdi for bioengineering and drainage management.
- **Landslide database:** A landslide database has been developed in coordination with National Housing Settlements and Resilience Platform (NHSRP) and respective provincial government. Stakeholders recognize database utility and have requested enhancements for decision support and institutionalization, aiming for data-driven decision-making across all sectors.

CRS Nepal, learning from GTS implementation approach, is now integrating **geohazard assessments** for site safety of shelter and settlements into other projects such as Jajarkot Earthquake Response Project (JERP), Lamjung Recovery Project (LRP), and Joint Recovery Action Plan (JRAP), enhancing resilience and recovery efforts.

86 assessment sites in **22** local municipalities across **5**

More than **2100** vulnerable families served

Time frame: 2023

Lessons

- Limited technical capacity at local governments with limited resources for mitigation, often prioritizing immediate relocation over structural mitigation measures.
- Ambiguity in DRM roles— especially for mitigation purposes. Resource requests from local governments gets diverted within ministries.
- Need for standardized reporting of geohazard assessment for timely and informed decision making.

Learning from Gorkha Landslide Response (GLR), a Start-Fund funded anticipatory project

The GLR project implemented in 2022 in Arughat Rural Municipality, Gorkha highlights the necessity of adopting an integrated approach that incorporates nature-based solutions (NbS) alongside engineering structures. Suggested measures included: a) plantation of walnut trees in upslope section; b) use of live check dams in conjunction with gabion check dams along gullies and engineering structures like concrete catch drains, gabion walls and diversion weir.

Debris flow at Gorkha (above) and typical structure of proposed Landslide/Debris flow protection works (below). Photo: CRS Nepal

Explaining the existing condition of the geohazard and the next steps to the vulnerable community members and local representatives. Photo: CRS Nepal

Recommendations

- **Strengthen the technical capacity of local government's technical staff** by providing training and orientation; a low-maintenance online certification course to municipal technical staff on geohazard assessment techniques and mitigation measures.
- **Strengthen government DRM systems** with availability of relevant technical expertise. This will allow development of technical strategies and knowledge uptake for cumulative progress.
- **Development of central database** that encompasses the needs assessment, hazard/risk information, and mitigation status of geohazard-related issues at the local government level which is crucial for enabling evidence-based decision-making.

Landslides at Odak, Taplejung, Koshi Province triggered after a torrential rainfall. GTS in coordination with NDRRMA/Start Network/NHSRP conducted a timely geohazard assessment suggesting permanent relocation of 63 households to safer location. Photo: CRS Nepal

CRS Nepal, Country Office
Sanepa- 02, Lalitpur Metropolitan City, Nepal
[CRS Nepal Site] | G.P.O. 8975-EPC 419 | [CRS Sanepa Office-Map]

For more information, contact Anish Khanal, Engineering Geologist/CRS Nepal.

Email: anish.khanal@crs.org